

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1345

Amelia St 100 kV
August 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1345. The Interconnection Customer proposes a 164.8 MW solar photovoltaic facility interconnecting at the Amelia St 100 kV substation in the Indiana area. The request was received on November 30, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 390 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,388,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Schuett - Sand Ridge 100 kV	Amelia St - Lansing 100 kV	227.9	115	100.0
Lansing - Amelia St 100 kV Line	Schuett - Sand Ridge 100 kV Line	201.5	130	100.0
Lansing Substation - Dyckesville 100 kV line	Castle Afb - Sand Ridge 100 kV	197.2	133	38.0
Sand Ridge - Castle Afb 100 kV	Dyckesville - Lansing 100 kV	175.4	134	33.8
Castle Afb - Shoreham Power Station 100 kV line	East Jackson 1&2 - Dyckesville 100 kV	135.1	137	26.7
Castle Afb Substation - Dyckesville 100 kV	Castle Afb - Sand Ridge 100 kV	122.9	135	11.6
Shoreham Power Sta - Outlook 100 kV	South Holland - Shoreham Power Sta 100 kV	118.5	118	16.7
East Jackson 1&2 - Dyckesville 100 kV	Castle Afb - Shoreham Power Station 100 kV Line	115.5	154	36.0
Sand Br - Wildwood 100 kV Cir. 2	Wildwood - Sand Bar 100 kV #1	105.3	144	6.2
Wildwood - Sand Bar 100 kV	Wildwood Substation - Sand Ar 100 kV Cir. 2	104.8	145	6.2
Robbins Community Power Llc - Sandbar 100 kV Line 1	Robbins Community Power Llc - Sand Bar 100 kV #2	104.4	145	6.0
South Holland - Shoreham Power Sta Substation 100 kV	Shoreham Power Station - Outlook 100 kV	100.0	131	14.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Schuett - Sand Ridge 100 kV	Rebuild line with higher-capacity conductor	\$1,489,000	55% (\$811,000)
Amelia St - Lansing 100 kV Line	Rebuild line with higher-capacity conductor	\$5,046,000	55% (\$2,748,000)
Lansing Substation - Dyckesville 100 kV line	Rebuild line with higher-capacity conductor	\$1,422,000	40% (\$564,000)
Sand Ridge - Castle Afb 100 kV	Rebuild line with higher-capacity conductor	\$1,125,000	40% (\$446,000)
Catle Afb - Shoreham Powe Station 100 kV	Rebuild line with higher-capacity conductor	\$2,170,000	37% (\$808,000)
Castle Afb Substation - Dyckesville 100 kV	Rebuild line with higher-capacity conductor	\$2,464,000	39% (\$956,000)
Shoreham Power Sta - Outlook 100 kV	Rebuild line with higher-capacity conductor	\$1,771,000	41% (\$730,000)
E Jackson 1&2 - Dyckesville 100 kV	Rebuild line with higher-capacity conductor	\$1,629,000	44% (\$710,000)
Wildwood - Sand Bar 100 kV Line 2	Reconductor line	\$505,000	28% (\$142,000)
Wildwood - Sand Bar 100 kV L	Reconductor line	\$570,000	28% (\$160,000)
Robbins Community Power Llc - Sandbar 100 kV Line 1	Reconductor line	\$904,000	28% (\$254,000)
S Holland - Shoreham Power Station 100 kV Line	Reconductor line	\$144,000	41% (\$59,000)
	Total allocated to project		\$8,388,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.